



GETTING STARTED WITH VERSION CONTROL: INTRO TO GIT AND GITHUB

Welcome!



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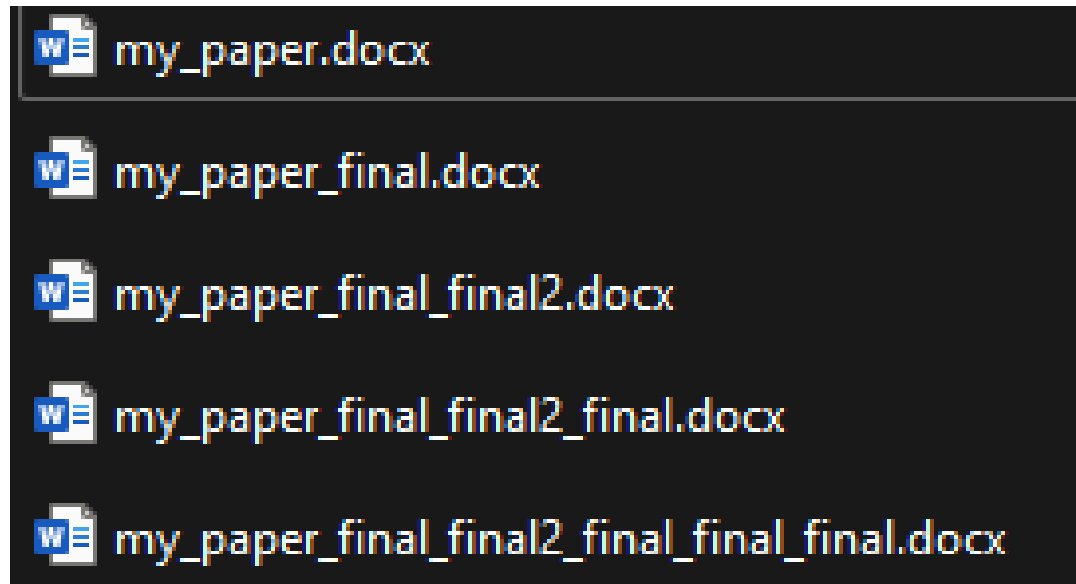
Plan for Today

- 14:00** Welcome & introduction
- 14:20** *Live Demo & troubleshooting:* Create your first repo & your first commit
- 14:40** *Live Demo & troubleshooting:* Create a new branch and merge it
- 15:00** Break
- 15:10** *Live Demo & troubleshooting:* Fork a repo & create a pull request
- 15:30** *Live Demo & troubleshooting:* Resolving a merge conflict
- 15:50** Wrap up & additional resources
- 16:00** End of workshop



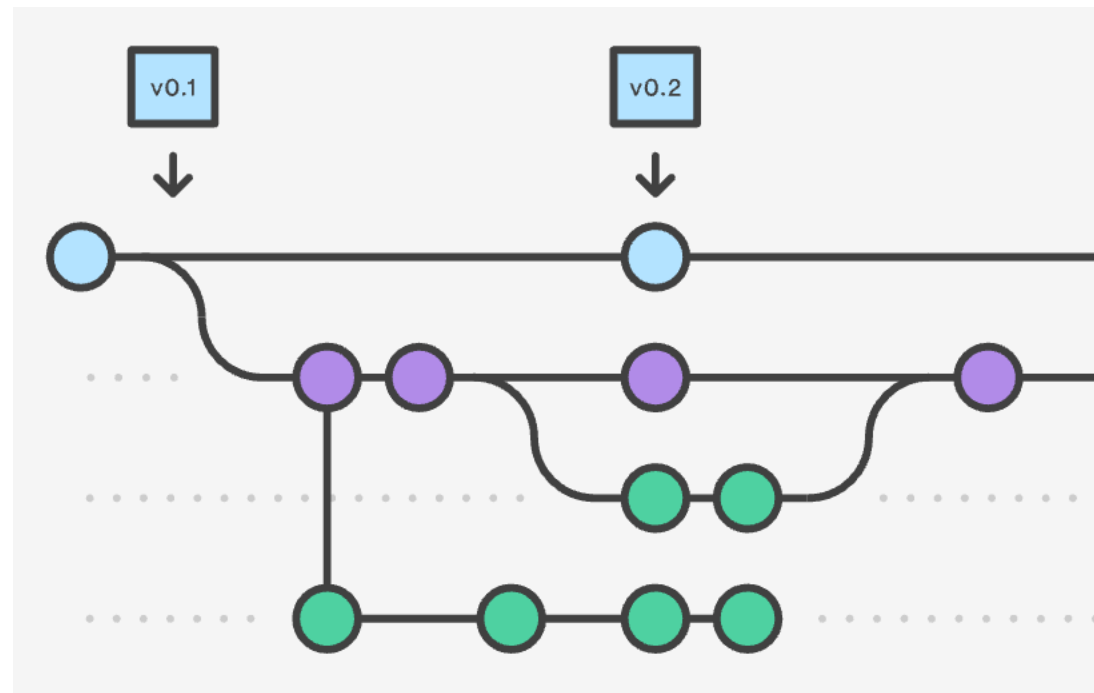
Version Control

- **Version control** helps you track changes to your files.
- You probably already have done (poor) version control before!



Version Control

- **Version control is the practice of tracking and managing *changes* to files** (traditionally, software code).
- Version control tracks every individual change by each contributor and helps prevent *concurrent* work from conflicting.
- It is a standard *workflow* in the tech industry.



git

Git is a version control system installed and maintained on your local system

```
singh@DESKTOP-PGVSHMF MINGW64 ~/Desk... (th)
$ git checkout master
Switched to branch 'master'

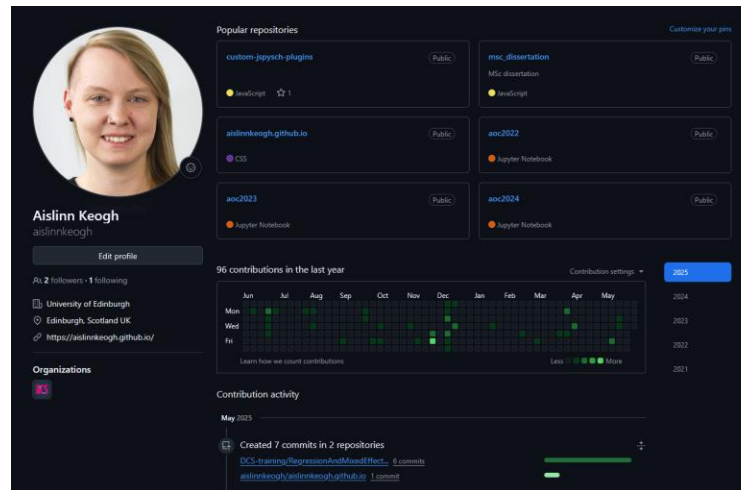
singh@DESKTOP-PGVSHMF MINGW64 ~/Desk... (th)
$ git merge newbranch
Updating 9e7f7d0..1a1b1b1
Fast-forward
 branch file.txt
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
```

No need to learn this for basic use cases!

Image from <https://funkyvast.weebly.com/what-is-git-bash-terminal.html>

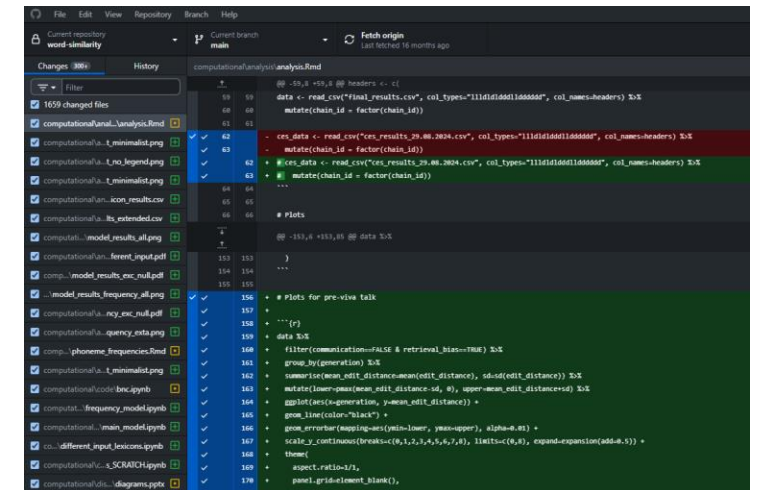
GitHub

GitHub is a hosting service for Git repositories. It is exclusively cloud-based



GitHub Desktop

A desktop application for working with git and GitHub on your computer



Terminology

repository/repo

commit

clone

branch

merge

fetch

pull

push

fork



Confused Person - ClipArt B...
clipartbest.com



Confused | Free Images at Clker...
clker.com



Confused Look Emoticon | ww...
imgkid.com



Confused Man Fre...
publicdomainpictur...



http://www.dreamstime.com/royalty-...
happyhormonesforlife.com



Confuse gesture, Confusion Medical s...
pngegg.com



Image Of Confused Person...
cliparts.co



Confused Cartoon ...
cliparts.co



Textusa: Perhaps... Confusing
textusa.blogspot.com



Confused person png imag...
lovepik.com



January | 2010 | Investing Caffeine
investingcaffeine.com



Principals Share The Most Bizarre ...
humaverse.com



Don't Worry, We're All Confused | Show...
showmeinstitute.org



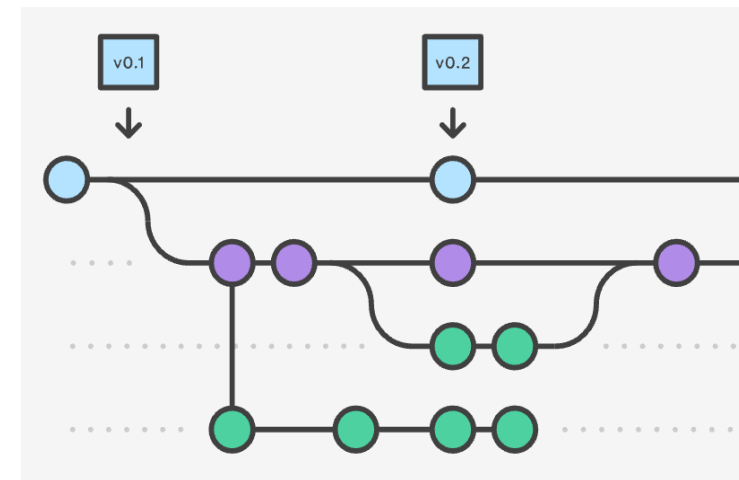
confused frustrated business man dreamst...
jobcrusher.com

Terminology

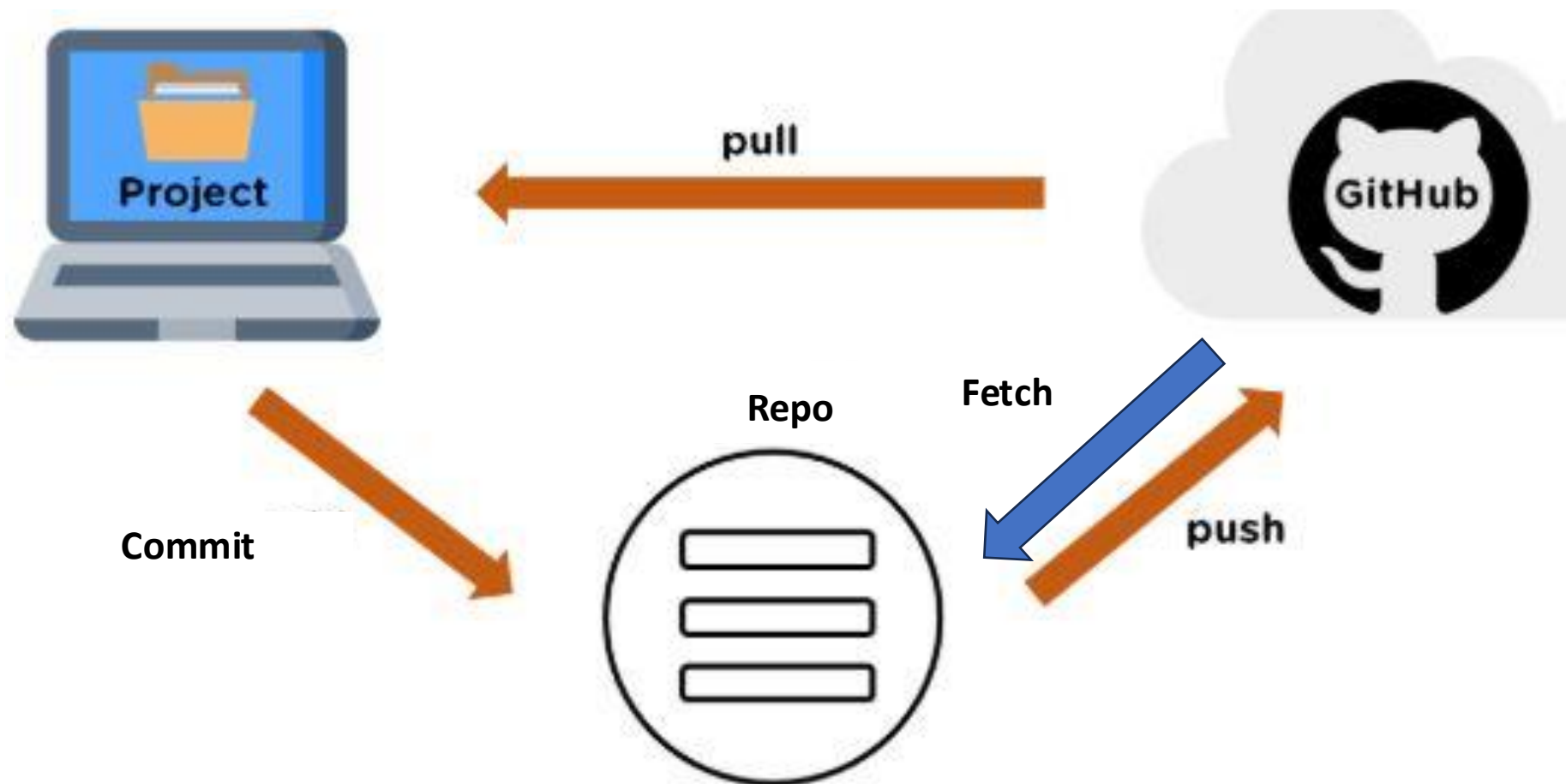
**repository/
repo** → a “folder” for your code for one project
It's a place where you can store your code,
your files, and each file's revision history.
Repositories can have multiple
collaborators and can be either public or
private.

clone → a copy of a repo (e.g. on your computer)

branch → a parallel version of a repo (repos can
have multiple branches)



Terminology



Creating your first repo

1. Go to www.github.com
2. Sign in / sign up
3. Click the + in the top right corner and select “New repository”
4. Choose a repository name, check “Add a README file”, and leave all other fields.
5. Click “Create repository”



Cloning your repo with GitHub Desktop

1. Install GitHub Desktop from <https://desktop.github.com> if you haven't yet
2. Open GitHub Desktop, and log in with your account.
3. Locate your Github repository and click “Clone [your repo name].”
4. Choose the location (local path) of your repo on your machine and proceed.



Making a commit

1. Open the README.md file in a text editor of your choice
If you're unsure: on Windows use **Notepad**; on Mac use **TextEdit**
 - If you can't find the README.md file in your repo folder, you can create a README.md file using the editor.
2. Make some changes in your README file and save them
3. In GitHub Desktop, describe your commit and click commit when ready
4. Push to GitHub



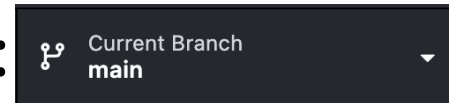
Terminology

repository/repo	→ a “folder” for your code for one project
clone	→ a copy of a repo (e.g. on your computer)
commit	→ saving your changes in the project
push	→ uploading your changes (your commits) to the server
fetch	→ downloading the latest changes from the remote repo
pull	→ integrating the latest changes into your clone of the repo
branch	→ a parallel version of a repo (repos can have multiple branches)
merge	→ combine the changes from one branch into another
fork	



Creating a new branch and merging it

1. In Github Desktop, open the branch menu:
2. Select 'new branch' and enter a name of your choice. Confirm that this branch is now displayed under 'Current Branch'.
3. Publish the new branch.
4. Make some changes in your README file. Save, commit, push.
5. Go to Branch -> Merge into the current branch and select your new branch. Create a merge commit, then push.



The background is a collage of three images. The top half features a black and white photograph of a room with several people working at desks, with a map of London on the wall. The bottom left features a red-tinted, high-contrast image of a woman in a floral dress holding a book. The bottom right features a black and white photograph of a man in a suit looking down at a desk.

Break

**GETTING STARTED WITH
VERSION CONTROL: INTRO
INTRO TO GIT AND GITHUB**

Terminology

- repository/repo** → a “folder” for your code for one project
- clone** → a copy of a repo (e.g. on your computer)
- commit** → saving your changes in the project
- push** → uploading your changes (your commits) to the server
- fetch** → downloading the latest changes from the remote repo
- pull** → integrating the latest changes into your clone of the repo
- branch** → a parallel version of a repo (repos can have multiple branches)
- merge** → combine the changes from one branch into another
- fork** → a personal copy of someone else’s repo



Live Demo:

Forking a repo, making a commit, and making a pull request

1. Find a partner and get the link to their GitHub repo. Fork it on GitHub, then add your fork to GitHub Desktop.
2. Make a change to their README, commit, and push.
3. On GitHub, view your fork and open a pull request.
4. Review and approve each other's pull requests.



Live Demo:

Resolving a merge conflict

1. Choose your or your partner's repo.
2. In your fork of your partner's repo, make changes to README, commit, and push to your own fork.
3. Your partner also makes changes to the README, commit, and push to their own repo.
4. Make a pull request for your partner's repo. This will create a merge conflict. Resolve as demonstrated.



Additional Resources

- GitHub online tutorials
 - Introduction (tutorial repeating some of what we did today): <https://github.com/skills/introduction-to-github>
 - Reviewing pull requests: <https://github.com/skills/review-pull-requests>
 - Resolving merge conflicts: <https://github.com/skills/resolve-merge-conflicts>
 - List of all skills tutorials: <https://github.com/skills>
- GitHub student pack: <https://education.github.com/pack>

